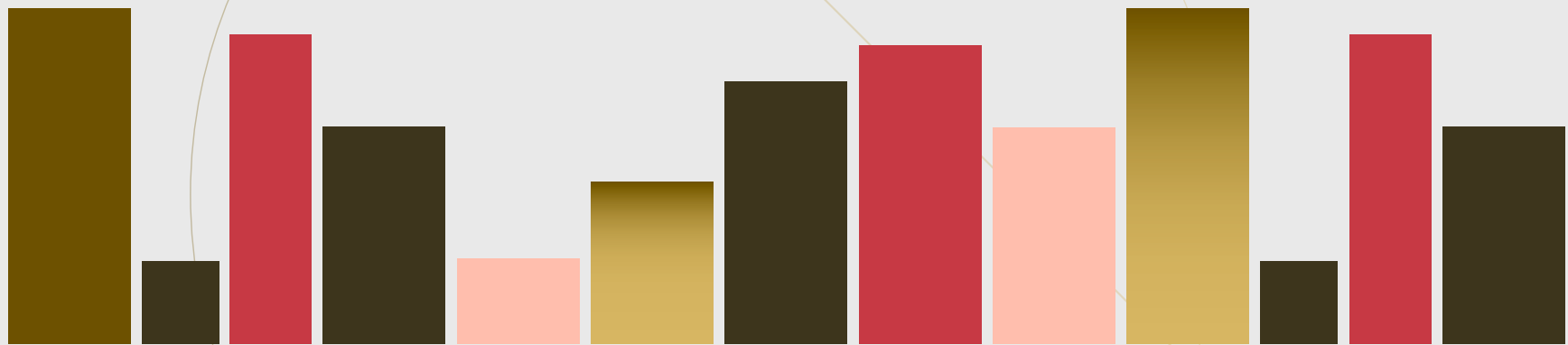


ZEN Univ. IUT-1



Exercise 5th.

$$1. \quad (a) \quad \begin{cases} x \equiv 8 & \text{mod } 28 \\ x \equiv 12 & \text{mod } 44 \end{cases}$$

$$\Leftrightarrow \begin{cases} x \equiv 0 & \text{mod } 4 \\ x \equiv 1 & \text{mod } 7 \\ x \equiv 1 & \text{mod } 11 \end{cases}$$

$$\begin{array}{ll} 7 \cdot 11 \equiv 1 & \text{mod } 4 \\ 4 \cdot 11 \equiv 2 & \text{mod } 7 \\ 4 \cdot 7 \equiv 6 & \text{mod } 11 \end{array} \quad \begin{array}{ll} 1^{-1} \equiv 1 & \text{mod } 4 \\ 2^{-1} \equiv 4 & \text{mod } 7 \\ 6^{-1} \equiv 2 & \text{mod } 11 \end{array}$$

By CRT, the solution mod $\text{lcm}(28, 44)$ is

$$0 \cdot 7 \cdot 11 \cdot 1 + 1 \cdot 4 \cdot 11 \cdot 4 + 1 \cdot 4 \cdot 7 \cdot 2 = 232$$

$$\text{lcm}(28, 44) = 308$$

$$\therefore 232 + 308k \quad (k \in \mathbb{Z})$$



(b)

$$\begin{cases} 8x \equiv 4 \pmod{13} \\ 16x \equiv 10 \pmod{21} \end{cases}$$

$$8^{-1} \equiv 5 \pmod{13}, \quad 16^{-1} \equiv 4 \pmod{21}.$$

$\Leftrightarrow$

$\gcd(8, 13) = 1$
 $\gcd(16, 21) = 1$

$$\begin{cases} x \equiv 7 \pmod{13} \\ x \equiv -2 \pmod{21} \end{cases}$$

$$13^{-1} \equiv 13 \pmod{21}, \quad 21^{-1} \equiv 5 \pmod{13}$$

By CRT, the solution mod $\text{lcm}(13, 21) = 273$,

$$7 \cdot 21 \cdot 5 + (-2) \cdot 13 \cdot 13 = 397 \equiv 124 \pmod{273},$$

$$\therefore 124 + 273k \quad (k \in \mathbb{Z}). \quad \textcircled{\text{Q.E.D.}}$$

$$2. \quad \phi(60) = 60 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 16$$

$$\left(7^{\underline{387}} + 11^{\underline{183}} + 13^{\underline{50}} \right)^{\underline{148}} \quad \left(\begin{array}{l} \text{mod } 60 \\ \text{for all} \end{array} \right)$$

mod $\phi(60)$ on exponents.

$$\equiv \left(7^{\underline{3}} + 11^{\underline{7}} + 13^{\underline{2}} \right)^{\underline{4}}$$

$$11^2 \equiv 1$$

$$\equiv (343 + 11 + 69)^4 \equiv (43 + 11 + 49)^4$$

$$\equiv 43^4 \equiv 1849^2 \equiv 49^2 \equiv 2401 \equiv 1$$

~~111~~

3. (a) $2 \nmid p$ so $2 \mid p-1$. if $\frac{p-1}{2} = 2$ then $p=5$, so $\frac{p-1}{2} \neq 2$.

$$\frac{p-1}{2} \leq p-2 \quad (\Leftrightarrow) \quad p-1 \leq 2p-4 \quad (\Leftrightarrow) \quad 3 \leq p \quad (\Leftrightarrow) \quad p > 5$$

Thus $(p-1) \mid (p-2)! = \underline{2 \cdot 3 \cdots \frac{p-1}{2} \cdots (p-2)}$ 

(b) $p^{n-1} + p^{n-2} + \cdots + 1 = \frac{p^n - 1}{p-1}$

Substitute p^n with $(p-1)! + 1$, we'll get

$$\frac{(p-1)!}{p-1} = (p-2)! \quad \text{this is divisible by } (p-1) \text{ by (a).}$$

$$\underline{0} \equiv p^{n-1} + p^{n-2} + \cdots + 1 \equiv 1^{n-1} + 1^{n-2} + \cdots + 1^0 \equiv \underline{n} \pmod{p-1} \quad \text{QED}$$

$$(c) \quad (p-1)! < (p-1)^{p-1} < p^{p-1}$$

$$\Rightarrow (p-1)! + 1 < p^{p-1}$$

$$\Rightarrow p^{p-1} - 1 > (p-1)!$$

n can't be ≤ 0 , so $n \geq p-1$ with (b), but this contradicts. 

$$4. \quad \exists e. n = 2^e \iff \underline{2\phi(n) \mid n}$$

Prf. $(\Rightarrow)$ (easy) $2\phi(n) = 2 \cdot \frac{n}{2} = n \mid n$

$(\Leftarrow)$

Take the contrapositive. $\forall e. n \neq 2^e$

Let n be $2^{e_1} p_2^{e_2} \dots p_k^{e_k}$ ($e_1 = 0$ if odd, $k \geq 2$)

$$2\phi(n) \mid n \implies \sqrt{2}(2\phi(n)) \leq \sqrt{2}(n)$$

so

$$\underline{2\phi(n) \nmid n} \iff \underline{\sqrt{2}(2\phi(n)) > \sqrt{2}(n)}$$

$$\begin{aligned}
& \sqrt{2}(2\phi(n)) \\
&= 1 + \sqrt{2} \left(n \prod_{i=1}^k \frac{p_i - 1}{p_i} \right) \\
&= 1 + \sqrt{2} \left(2^{e_1} (p_2 - 1) \left(\begin{cases} \frac{1}{2} & (\text{if even}) \\ 1 & (\text{if odd}) \end{cases} \right) (\dots) \right) \\
&\geq 1 + e_1 + 1 + (-1)
\end{aligned}$$

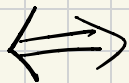
include $p_3, \dots$ and
 $p_2^{e_2} p_3^{e_3} \dots$
 $\downarrow$

$$\underline{\sqrt{2}(2\phi(n))} \geq e_1 + 1 > e_1 = \underline{\sqrt{2}(n)} \quad \blacksquare$$

② $\sqrt{2}(n)$ is exponential number of n

5.

$$\exists b. m \equiv b^n \pmod{p}$$



$$m^{\frac{p-1}{d}} \equiv 1 \pmod{p}$$

where $d = \gcd(p-1, n)$

Prf. ($\Rightarrow$) (easy). Let $n = ds$,

$$m_d^{\frac{p-1}{d}} \equiv b^{n \cdot \frac{p-1}{d}} \equiv b^{s(p-1)} \equiv b^{0 \cdot s} \equiv 1 \pmod{p}$$

Fermat's Little Theorem
FLT

($\Leftarrow$). $\mathbb{Z}/p\mathbb{Z}$ should have some primitive root g .

$$\text{Let } g^k \equiv m \pmod{p}.$$

$$\text{Assume } b^n \equiv m \pmod{p},$$

$$\text{Let } b \equiv g^x \pmod{p}$$

$$g^{xn} \equiv g^k \pmod{p}$$

so $k = xn + u(p-1)$ for some u . by FLT.

This is equivalent to $k \mid \gcd(p-1, u) = d$
by Euclidian method.

Now, we have $m^{\frac{p-1}{d}} \equiv 1$.

$$g^{k \frac{p-1}{d}} \equiv 1 \quad \text{so} \quad k \frac{p-1}{d} = v(p-1) \quad \text{for some } v.$$

$$\frac{k}{d} = v \quad \text{and this means} \quad \underline{k \mid d} \quad \#$$